Intelligent Manufacturing Quality Control

A working local visual-inspection and model-lifecycle platform, designed for reviewability rather than autonomous industrial decisions.

Context

The project explores the engineering of image classification and operational governance: trained models, ONNX inference, inspection history, human review and controlled model release. It is an independently built portfolio system, not a client deployment.

Objectives

- Classify six defined steel-surface defect categories in a reproducible local evaluation flow.
- Preserve individual predictions and human corrections as separate traceable evidence.
- Provide model evaluation, activation, rejection and rollback operations.

Scope of work

- Frontend: React, Vite and TypeScript interfaces for Overview, Inspections, Review Queue, Model Lab and MLOps monitoring.
- Inference: Rust/Axum service running versioned ONNX artifacts through ONNX Runtime.
- Model development: TensorFlow training, stratified splits, evaluation, checkpointing and ONNX export validation.
- Data: Firestore records and local/GCS-compatible object-storage integration.
- Governance: model registry, quality gates, activation, rollback, monitoring snapshots and review history.

Intelligent Manufacturing Quality Control / Scope continued

Expected or implemented deliverables

- Implemented local application and API workflows with synthetic verification fixtures.
- Stored inspection results, reviewer corrections and versioned model evaluation records.
- Tested application, backend and model-integrity flows documented in the repository.
- Optional cloud deployment preparation, with explicit remaining operational prerequisites.

Boundaries and qualifications

- Not an industrial pass/fail system and not verified to detect arbitrary defects or normal products.
- Published synthetic-test scores are not industrial accuracy claims.
- Cloud production deployment and real-dataset qualification are not represented as completed.

Document basis

Source: github.com/keenosmith-del/intelligent-manufacturing-qc (project README and linked engineering documentation). The repository describes implemented local flows, synthetic verification and remaining limitations.

This document describes a project scope, not a contractual quote, client endorsement or guarantee of results. Commercial delivery, where relevant, requires a separately agreed statement of work.

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